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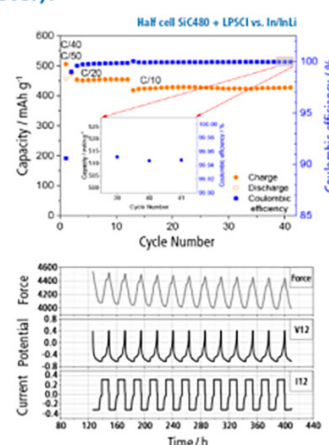
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Pore Network Modelling of Galvanostatic Discharge Behaviour of Lithium-Ion Battery Cathodes

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The performance of Lithium-Ion batteries (LIB's) strongly depends on 3D microstructure and continued research is needed for the development and optimization of electrode designs to further reduce cost and improve performance and durability. In this work, a pore network modelling approach is presented to understand the structure-performance relationship of porous cathodes of LIB's. It was demonstrated that pore network models can efficiently predict the rate-dependent capacity of an electrode using only a 3-phase tomogram as input. The developed modelling framework was used to perform structural analysis on two $\text{Li}(\text{Ni}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2})\text{O}_2$ (NMC532) cathodes of different thickness and calendaring pressure and revealed important insights of microstructural heterogeneities inside porous structures, including spatial distribution of concentration, potential and state of lithiation in electrolyte, active material and carbon binder domain. The computational performance of the pore network model was analyzed, and excellent performance was demonstrated, taking hours instead of weeks for a similar direct numerical simulation. The novel modelling framework reported in this study will enable the study of local heterogeneities in other types of cathode material to help screen next-generation electrode designs, augmenting and informing time-consuming cell fabrication and laboratory testing.

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Lithium-Ion Batteries (LIB) are currently the best available power source for many applications ranging from consumer electronics to electric vehicles due to their high energy and power density as compared to other energy storage devices.¹ However, increasing energy density, lowering cost, and extending life remains essential for broader adoption, especially in transportation. Many options for cathode materials are available, but lithium iron phosphate (LFP) and lithium manganese nickel cobalt oxide (NMC) are the most widely used commercially because of their relative high capacity and reduced price.^{2–4} Beside material selection, optimization of electrode microstructure can greatly enhance the transport and reaction processes and thus improve the overall performance of LIB's. For this purpose mathematical modelling can play a vital role to understand the structural—performance relationship of porous electrodes.^{5,6}

A multitude of mathematical models have been developed for investigating physical and electrochemical processes of LIB's^{7–10} however, most of them either consider the porous electrodes as homogenous structures (volume averaged) or incorporated structural heterogeneities using empirical relationships. The most widely used model is the “pseudo two dimensional” (P2D) based on porous electrode and concentrated solution theory developed by Newman and co-workers.^{8–11} Their model simulates the intercalation/deintercalation process by assuming active material as isotropic spherical particles of uniform size, and estimates the effective transport properties using the Bruggeman expression whose accuracy for non-spherical geometries is still controversial.¹² Although the simplifications of the P2D model do not capture the real microstructure, it is still the standard modeling approach due to ease of use. On the other hand, attempts have been made to extend the capabilities of P2D model by incorporating microstructural effects, but due to additional complexity the computational cost increases many fold, becoming infeasible to model large domains with normal computational resources.^{13,14} To gain a better understand of the impact of microstructure it is necessary to develop a computationally

efficient framework that captures the structural heterogeneities of porous components of LIB cell.

A promising option is pore network modelling (PNM), which is a computationally efficient technique that can simulate transport at the pore-scale, preserving structural heterogeneities of porous material,^{15,16} at low computational cost. The technique uses a simplified geometric representation of real porous structures by treating the pore space as a network, where cavities or pores are the nodes, and they are interconnected by constrictions or throats. Each pore is considered as a well-mixed body that exchanges species with its neighbouring pore bodies via network connections. These networks can be extracted from X-ray tomography (XCT) images of real porous materials via different types of network extraction algorithms.^{17–19} Recently, PNM models have shown their utility to simulate electrochemical devices like fuel cells¹⁹ and redox flow batteries¹⁹ but to the best of our knowledge, they have not been applied to the active materials of lithium-ion battery during charging and/or discharging. PNMs are ideally suited for such microstructural modelling since the computational efficiency compensates for the high demands of transient multiphysics modelling. Their lack of adoption is a least partially due to the lack of an existing PNM code-base capable of handling the various complexities of the problem, including the intercalation reaction, migration effects, and three intermixed phases (void, active material and carbon binder) that all must be captured.

PNMs have been successfully applied to electrochemical devices, and a few have looked specifically at Li-based batteries. Torayev et al.²⁰ developed a pore network model of super P carbon electrode of lithium-oxygen battery. The 3D network of pore phase was extracted to find the discharge behaviour of four different zones of same electrode tomography image. The results showed deviations of model results with experimental data due to the variation of pore interconnectivity in different zones. The model, however, does not incorporate the solid phase structural-performance relationship. Lagadec et al.²¹ highlighted the importance of pore interconnectivity in lithium-ion battery separator by comparing two different separators of the same porosity and tortuosity but different pore interconnectivity. The results showed that concentration gradient due to diffusion for both separators varied significantly despite the similar macroscopic properties. Khan et al.^{22–24} developed a 3D pore network model of lithium-ion battery cathode from a full resolved XCT image of $\text{Li}(\text{Ni}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1})\text{O}_2$ (NMC811). The model highlighted the impact

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Table I. Properties of XCT images of NMC532 cathodes.

Property	Units	NMC532 (1CAL)	NMC532 (3CAL)
Coating Thickness	μm	129	88
Volume Fraction	%	—	—
Electrolyte phase	—	36.8	36.6
Active Material (NMC532)	—	49.28	49.74
Carbon and Binder Domain (CBD)	—	13.92	13.66
Density	mg/cm^{-2}	29.78	20.40
Experimental C-rate	$\text{mAh}/\text{g}_{\text{NMC}}$	178	182
Image size	voxels	$320 \times 250 \times 250$	$221 \times 250 \times 250$
Voxel Size	nm	398	398
Extracted Network	(Nodes, Bonds)	4637, 31427	3510, 23126
Electrolyte phase	—	1238, 1943	982, 1459
Active Material (NMC532)	—	1722, 3736	1295, 2760
Carbon and Binder Domain (CBD)	—	1680, 3375	1233, 2310
Interconnections	Bonds	22373	16597
NMC532-CBD	—	7620	5355
Electrolyte-CBD	—	6602	4979
Electrolyte-NMC532	—	8151	6263

of treating carbon binder as separate phase. They used a PNM model to simulate steady-state diffusion-reaction and diffusion conduction process in electrolyte and solid-phase respectively but did not analyze transient intercalation/deintercalation process.

The alternative to pore network modeling is direct numerical simulation (DNS), where either (a) the voxels of a tomography image are treated as the mesh elements for a CFD solver, or (b) a conformal mesh is created to represent the void space. Several recent publications have attempted to model the impact of microstructure using DNS, despite the computational challenges. Chouchane et al.²⁵ simulated a 4D electrochemical model of LIB cathode using direct numerical simulation to investigate the spatial location of carbon binder domain on overall electrochemical performance of the battery. The results showed changes in overall cell performance due to spatial variation in carbon binder domain (CBD), and that CBD without nanopores can lead to decrease in current density by increasing the tortuosity of Li transport. Lu et al.²⁶ developed a fully resolved 3D microstructure model of NMC cathodes using XCT images and studied the extent to which that LIB performance is dependent on structural heterogeneities. They altered the particle size distribution to predict next-generation electrodes architecture. They did not report on pore-to-pore analysis of concentration and potential distribution, which is the main output of a pore scale model, and moreover, they used relatively small domain for the simulation. Clearly there is a crucial structure-performance dependence of porous cathode, but there is a need to develop a modelling framework that can simulate substantially larger porous domains without requiring special computational resources.

In the present work, an existing pore network modelling framework (OpenPNM) was extended to study the galvanostatic discharge behaviour of lithium-ion battery half-cell. The model incorporated the three-phase network of $\text{Li}(\text{Ni}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2})\text{O}_2$ (NMC532) cathode material extracted from X-ray tomography images, using a multiphase network extraction technique recently developed by the authors.^{22,27,28} The galvanostatic discharge behaviour of the two cathodes was then compared with experimental data and found to be in excellent agreement. Impressively, the PNM was able to predict capacity of both electrodes at various C-rates (between 0.2 and 3), using only the tomographic imaging data to generate the model. Moreover, the excellent computational efficiency of the PNM approach allowed modeling of relatively large domains, spanning the full thickness of a normal electrode, enabling it to capture the effect of structural heterogeneities in material. To the best of our knowledge, this is first transient pore network model of lithium-ion battery discharge and opens up a new avenue to study and optimize next-generation battery electrodes microstructures.

Modelling Approach

Cathode samples.—X-ray tomography images of two porous cathodes were obtained from open-source electrodes library of National Renewable Energy Laboratory (NREL).²⁹ Both cathodes consist of $\text{Li}(\text{Ni}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2})\text{O}_2$ (NMC532, TODA America Inc.) as active material, C45 carbon (Timcal), polyvinylidene fluoride (PVDF) (Solvay, Solef 5130) as carbon and binder domain and 1.2 M LiPF_6 in ethyl carbonate/ethyl methyl carbonate (EC:EMC, 3:7 w/w, Tomiyama, Japan) as electrolyte. The thickness of both electrodes was achieved by a multi-pass calendaring process with a constant but untracked applied force. The specifications of the two electrodes are listed in Table I. Segmented cathodes image containing two labels (0 and 1) were taken for analysis. Label 0 corresponds to the electrolyte and carbon binder domain (CBD) while label 1 corresponds to the NMC532 active material particles. The CBD phase was added using image morphology functions.³⁰ This technique was recently shown to be more suitable than other physics-based carbon binder models^{31–33} as it only binds the contact points between active material particles. A detailed comparison can be seen in Hein et al.³⁴ The cathode domain sizes used in this study are larger than the representative element volume (REV) that was estimated to be $50 \times 50 \times 50 \mu\text{m}^3$.³⁵ The renderings of both cathodes after addition of CBD phase can be seen in Figs. 1a, 1i. Importantly, experimental discharge curves are available for all formulations described above.

Pore network extraction.—The key to the following pore network modelling simulations presented herein was the extraction of three-phase networks from the tomograms. A watershed-based network extraction tool reported previously¹⁷ was adapted to work on multiphase images,²⁷ and it was confirmed that such networks could predict the effective transport properties in both the solid and void phase of NMC microstructures by comparing with direct numerical simulations.²² Crucially, the extracted network contains interconnections between each pair of phases, which is necessary to simulate interphase mass transport and Li^+ ions intercalation between the electrolyte and NMC nodes. The extracted networks for both cathodes samples can be seen in Figs. 1b, 1j where nodes on each network are shown as spherical balls while throats are shown as cylindrical sticks. The interconnections between phases are shown in Figs. 1c, 1k.

Governing equations.—The developed model simulates galvanostatic discharge behaviour of $\text{Li} - \text{Li}(\text{Ni}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2})\text{O}_2$ half-cell sandwich as shown in Fig. 2. The cell contains three regions: namely lithium foil as anode, porous separator filled with electrolyte

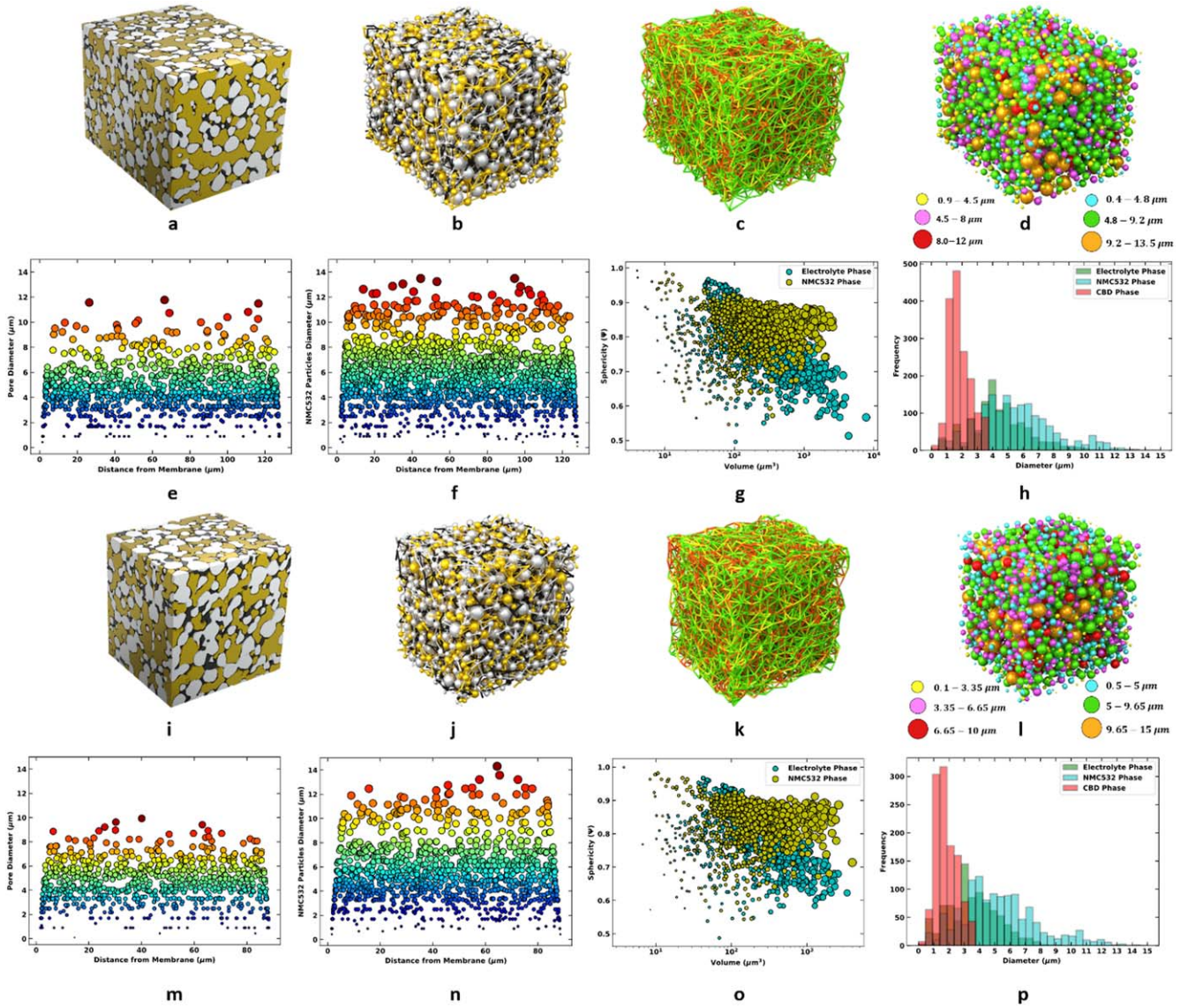


Figure 1. (a), (i) Three-phase rendering of 1CAL and 3CAL cathodes with golden, silver and black colour representing electrolyte, active material and CBD phase respectively, (b), (j) Extracted networks of 1CAL and 3CAL respectively, (c), (k) throat interconnection between two phases, green, yellow and red colour show interconnections between electrolyte-NMC532, electrolyte-CBD, NMC532-CBD respectively, (d), (l) Size distribution of electrolyte and NMC532 phase. The sizes are classified into three categories. Yellow, pink and red colour represent electrolyte phase size distribution while cyan, green and copper represent particle size distribution in NMC532 phase, (e), (m) Pore size distribution in the electrolyte phase of 1CAL and 3CAL with respect to distance from the separator, (f), (n) Particle size distribution in NMC532 phase for 1CAL and 3CAL with respect to distance from the separator, (g), (o) Sphericity vs Volume of pores particles in electrolyte and NMC532 phase, (h), (p) Size Distribution comparison of all three phases in 1CAL and 3CAL.

and porous cathode containing electrolyte, active material (NMC532) and carbon binder domain (CBD). During the discharge process lithium-ions are liberated from lithium foil anode, pass through the separator, and then into the cathode via the electrolyte-filled void; while electrons travel from the Li-foil anode to the cathode current collector via an external circuit. No expansion of active material upon insertion of lithium metal was considered to reduce model complexity.

The electrochemical reaction at the electrolyte/active material surface of cathode during galvanostatic discharge is:



The concentration (c) of the electrolyte in the cathode region can be estimated using the Nernst-Planck equation. The equation is written in discrete form for every i^{th} pore in electrolyte phase of pore network with k neighbors in the electrolyte phase as:

$$v_i \left(\frac{c_i^{t+\Delta t} - c_i^t}{\Delta t} \right) = \sum_{j=1}^{k_{elec}} [G_{ij}^d + \max(-G_{ij}^m(\varphi_i^t - \varphi_j^t), 0)] c_i^t - \sum_{j=1}^{k_{elec}} [G_{ij}^d + \max(G_{ij}^m(\varphi_i^t - \varphi_j^t), 0)] c_j^t + a_{j,n,rp}^t \quad [2.2]$$

where v_i is volume of pores, φ is electrolyte potential, a is the interfacial area between active material particle i and electrolyte pore j , and $j_{n,rp}$ is the pore wall flux of lithium ion across the interface between active material particle i and electrolyte pore j . This term is only applicable for reacting pores (rp) and is considered zero for pores that are not at the interface of electrolyte/active material particle. Lastly, G_{ij}^d and G_{ij}^m are the diffusive and migrative conductance between pore i and j that are calculated using the resistors-in-series analogy according to Eqs. 2.3 and 2.4 respectively:

$$G_{ij}^d = \left(\frac{1}{g_i^d} + \frac{1}{g_{ij}^d} + \frac{1}{g_j^d} \right)^{-1} \quad [2.3]$$

$$G_{ij}^m = \left(\frac{1}{g_i^m} + \frac{1}{g_{ij}^m} + \frac{1}{g_j^m} \right)^{-1} \quad [2.4]$$

where:

$$g_i^d = \frac{A_i D_{Li^+}}{l_i} \quad [2.5]$$

$$g_i^m = \frac{zF}{RT} g_i^d \quad [2.6]$$

The potential distribution in the electrolyte phase is calculated with respect to lithium reference electrode, which leads to the following potential equation written in terms of current density in the cathode region for every i^{th} pore with k neighbors in the electrolyte phase as:

$$i_{i,elec} = -zF \left(\sum_{j=1}^{k_{elec}} G_{ij}^d (c_i^t - c_j^t) \right) - \sum_{j=1}^{k_{elec}} K_{ij}^{elec} (\varphi_i^t - \varphi_j^t) \quad [2.7]$$

where $i_{i,elec}$ is the current density of i^{th} pore in the electrolyte phase of cathode region and K_{ij}^{elec} is the ionic conductance of the electrolyte solution between pore i and pore j as follows:

$$K_{ij}^{elec} = \left(\frac{1}{k_i^{elec}} + \frac{1}{k_{ij}^{elec}} + \frac{1}{k_j^{elec}} \right)^{-1} \quad [2.8]$$

where

$$k_i^{elec} = zF g_i^m c_i^t \quad [2.9]$$

where diffusivity and conductivity of Li^+ ion in the separator and cathode region is assumed to be dependent on electrolyte phase concentration. Also, lithium diffusivity in active material is assumed to be dependent on state of charge. The relationship to estimate diffusivity and conductivity is defined in Table II. The diffusion of lithium inside the active material was assumed to obey Fick's law and expressed for active material particle i with k neighbors in the active material phase (AM) for time step Δt as:

$$v_{i,AM} \left(\frac{c_{i,Li}^{t+\Delta t} - c_{i,Li}^t}{\Delta t} \right) = \sum_{j=1}^{k_{AM}} G_{ij}^{sd} (c_{i,Li}^t - c_{j,Li}^t) \quad [2.10]$$

where $v_{i,AM}$ is the volume of i^{th} particle of active material, $c_{i,Li}$ is the concentration of lithium ion of i^{th} active material particle and G_{ij}^{sd} is the diffusive conductance of lithium in the active material phase defined by:

$$G_{ij}^{sd} = \left(\frac{1}{g_i^{sd}} + \frac{1}{g_{ij}^{sd}} + \frac{1}{g_j^{sd}} \right)^{-1} \quad [2.11]$$

and

$$g_{ij}^{sd} = \frac{A_i D_{Li}}{l_i} \quad [2.12]$$

and v_i is the volume of the active material particle, having the same meaning as pore volume in Eq. 2.2.

The distribution of potential of an arbitrary particle i in the active material and carbon binder phase of cathode region was calculated using Ohm's law by Eqs. 2.13 and 2.14 respectively. It was assumed that no reaction takes place in carbon binder phase.

$$i_{i,AM} = - \sum_{j=1}^{k_{AM}} G_{ij}^{AM} (\varphi_{i,AM}^t - \varphi_{j,AM}^t) \quad [2.13]$$

$$0 = - \sum_{j=1}^{k_{CBD}} G_{ij}^{CBD} (\varphi_{i,CBD}^t - \varphi_{j,CBD}^t) \quad [2.14]$$

where G_{ij}^{AM} and G_{ij}^{CBD} are electrical conductance in active material and carbon binder domain respectively. The values are estimated using similar application of the resistors-in-series analogy as previously defined in Eq. 2.8. The value of conductance in arbitrary active material or carbon binder particle i is defined as:

$$g_{ij}^e = \frac{A_i \sigma_{AM/CBD}}{l_i} \quad [2.15]$$

The Butler-Volmer expression was used to relate phase potentials to reaction kinetics of the intercalation process⁸ of an arbitrary pore i as shown by Eq. 2.16 below. The exchange current density, overpotential and equilibrium potential polynomial fit³⁸ were defined according to Eqs. 2.17, 2.18, and 2.19 respectively.

$$a_{j,n,i}^{j,t} = \frac{\nabla i_{elec}}{zF} = i_i^{0,t} \left(\exp \left(\frac{(1 - \alpha_c)F}{RT} \eta_{i,c}^t \right) - \exp \left(\frac{-\alpha_c F}{RT} \eta_{i,c}^t \right) \right) \quad [2.16]$$

$$i_i^0 = \sum_{j=1}^{k_{elec}} (a_{ij} F k (c_{j,elec}^{1-\alpha_c})' (c_{i,max}^t - c_{i,AM}^t)^{1-\alpha_c} (c_{i,AM}^t)') \quad [2.17]$$

$$\eta_{i,c}^t = \sum_{j=1}^{k_{elec}} \varphi_{i,AM}^t - \varphi_{j,elec}^t - U_{i,cathode}^t \quad [2.18]$$

$$\begin{aligned} U_{i,cathode} = & 5744.862289 \text{ soc}_i^9 - 5520.41099 \text{ soc}_i^8 \\ & + 95714.29862 \text{ soc}_i^7 - 147364.5514 \text{ soc}_i^6 \\ & + 142718.3782 \text{ soc}_i^5 - 90095.81521 \text{ soc}_i^4 \\ & + 37061.41195 \text{ soc}_i^3 - 9578.599274 \text{ soc}_i^2 \\ & + 1409.309503 \text{ soc}_i - 85.31153081 \\ & - 0.0003 \exp(7.657(\text{soc}_i^{115})) \end{aligned} \quad [2.19]$$

The current in the solid and electrolyte phase is conserved through:

$$\nabla i_{elec} + \nabla i_{AM} = 0 \quad [2.20]$$

The concentration and potential gradient in the separator region of electrolyte phase were determined using Eqs. 2.21, 2.22 with boundary condition for lithium ion dissociation at anode/separator interface represented by Eq. 2.23.

$$\frac{\partial c_{2,s}}{\partial t} = -D_{Li,s}^{eff} \nabla^2 c_{2,s} \quad [2.21]$$

$$\frac{\partial \varphi_{2,s}}{\partial x} = -\frac{i_2}{\kappa_s^{eff}} + \frac{RT}{F} (1 - t_+) \frac{\partial \ln c_{2,s}}{\partial x} \quad [2.22]$$

$$\frac{\partial c_{2,s}}{\partial x} = \frac{I(1 - t_+)}{F D_{Li,s}} \quad [2.23]$$

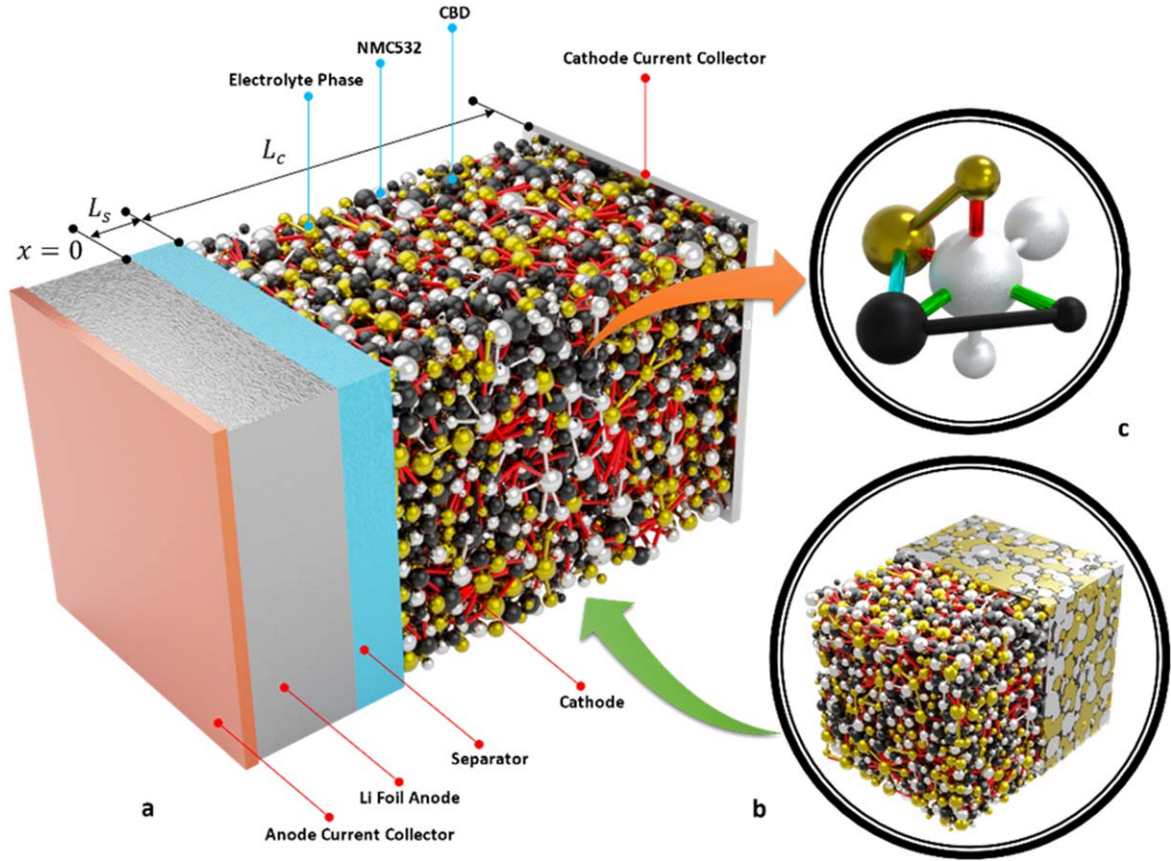


Figure 2. (a) Lithium Ion battery Half cell pore network modelling schematic, (b) Overlaid extracted network from XCT image of cathode, (c) Unit cell showing different phases connection with each other. Golden, silver and black colour represent electrolyte, NMC532 and carbon black nodes and single-phase interconnection while the red, green and cyan colour represents interconnection between two different phases.

Table II. Lithium-Ion battery half cell PNM model parameters.

Parameter	Symbol	Units	Value
Separator Thickness	L_s	μm	25
Separator Porosity	ε_{sep}	%	39
Electronic conductivity of NMC532 ²⁵	σ_{AM}	S/m	0.01
Electronic conductivity of CBD ³⁶	σ_{CBD}	S/m	760
Ionic conductivity of the electrolyte in the separator ³⁷	K_s	S/m	(1)
Initial Li^+ concentration in the electrolyte ²⁹	$c_{2,0}$	mol/m^3	1200
Maximal Li concentration in NMC532 ²⁹	$c_{1,\text{max}}$	mol/m^3	48900
Diffusion coefficient of Li^+ in the electrolyte ³⁷	D_{Li^+}	m^2/s	$10^{-4.43 - \frac{54.0}{T-229-5c_2} - 0.22c_2}$
Li diffusion coefficient in AM ²⁹	D_{Li}	m^2/s	(2)
Faraday constant	F	C/mol	96485
Cathode rate constant	k	$\text{mol}/\text{m}^2\cdot\text{s}$	1e-10
Bruggeman exponent coefficient for the separator	p_{sep}	—	1.5
Ideal gas constant	R	J/mol.K	8.314
Reference temperature	T	K	303
Li transference number in separator	t_+	—	0.363
Cut off Voltage	V	V	3.0
Charge transfer coefficient of NMC	α_c	—	0.5
Charge transfer coefficient of Li foil	α_a	—	0.5
Exchange current density of Li foil ²⁹	i_{foil}^0	A/m^2	19

(1) $10^{-2319\text{soc}^{10} + 6642\text{soc}^9 - 5269\text{soc}^8 - 3319\text{soc}^7 + 10038\text{soc}^6 - 9806\text{soc}^5 + 5817\text{soc}^4 - 2286\text{soc}^3 + 575.3\text{soc}^2 - 83.16\text{soc} - 9.292}$.

(2) $c_2(-10.5 + 0.0740T - 6.96e^{-5}T^2 + 0.668c_2 - 0.0178c_2T + 2.80e^{-5}c_2T^2 + 0.494c_2^2 - 8.86e^{-4}c_2^2T^2)$.

The electrolyte phase potential drop at the interface of anode/separator was determined using the Butler Volmer expression at the Li foil anode region:

$$\nabla i_{elec} = i_{foil}^0 \left(\exp \left(\frac{(1 - \alpha_a)F}{RT} (\varphi_{Li} - \varphi_{2,s}) \right) - \exp \left(\frac{-\alpha_a F}{RT} (\varphi_{Li} - \varphi_{2,s}) \right) \right) \quad [2.24]$$

The overall cell potential was defined as the difference of potential between the anode and cathode current collector. This study only focuses on the three-phase pore network model of cathode region in half cell sandwich, therefore, separator and lithium foil regions were included using 1D modelling equations to simulate boundary conditions at separator/cathode boundary region and ultimately estimate cell potential during the galvanostatic discharge process. The solution workflow is described in more detail below.

Boundary conditions.—The boundary conditions implemented in the electrolyte, active material and carbon binder phase in half cell are as follows:

1. In the *cathode region*, no flux boundary conditions for current and lithium ion transport were set in the electrolyte phase at $x = L_s + L_c$. Similarly, no flux boundary conditions of lithium in the active material phase were set at active material/separator interface and active material/ cathode current collector interface. For electronic flux in the solid phase, a no flux boundary was set at the interface of separator/solid phase (CBD + AM) at $x = L_s$. Lastly, continuity of mass, mass flux, charge and charge flux were enforced at separator/electrolyte phase interface at $x = L_s$.
2. In the *separator region*, a no flux current boundary condition was set at anode/separator interface at $x = 0$. Also, the potential of the lithium solid phase (φ_{Li}) was arbitrarily set to zero at this interface. Lastly, the flux of lithium ions was calculated by specifying the boundary condition using Eq. 2.23.

Workflow and validation.—The pore network modelling workflow is shown in Fig. 3. To simulate the half-cell, all parameters, material transport properties, boundary and initial values were specified in the first step. The values of all parameters are reported in Table II. Based on the initial and boundary conditions at a specific current density, the concentration profile in the electrolyte phase of the separator and cathode region was simulated. For this purpose, a 1D model of diffusion in the separator was coupled with the pore network model of the porous cathode at separator/cathode interface by applying continuity and constant flux boundary conditions. The 1D model in the separator region is necessary as concentration of lithium changes at separator/cathode interface with time of discharge, therefore a fixed concentration boundary condition at this interface was not suitable.^{39,40} Following this approach, a 1D model of potential distribution was solved in the separator to calculate a boundary value for the potential at the separator/cathode interface after calculating concentration distribution. This allowed estimation of the potential distribution in the electrolyte phase of the cathode region. This was achieved by applying Eqs. 2.23 and 2.7 in the separator and cathode, respectively. The distribution of concentration and potential in electrolyte phase pores allowed simulating the li-ion insertion reaction at electrolyte/active material interface according to Butler Volmer expression defined in Eqs. 2.16, 2.17, 2.18. The Butler Volmer expression requires the active material (NMC532) concentration of lithium and potential distribution in the solid phase of the cathode region. The concentration and potential distribution in the solid-phase were also affected by carbon binder domain that is spatially distributed to enhance electronic conductivity of the cathode material. Therefore, to achieve the desired

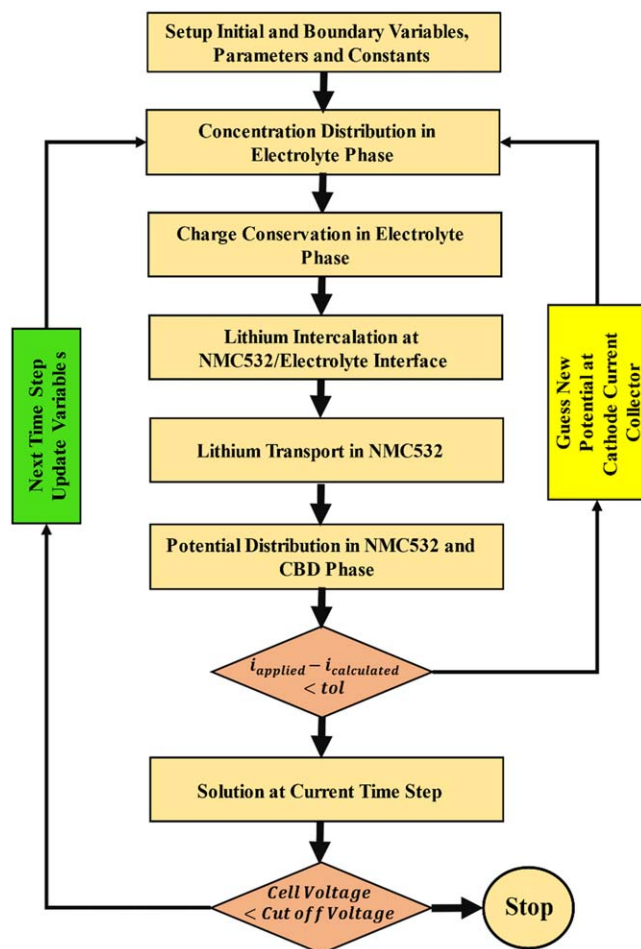


Figure 3. Pore network modelling workflow for galvanostatic discharge.

current density at the cathode current collector, an iterative process was adopted. For this purpose, a potential at the cathode current collector was guessed and the potential distribution in NMC532 and carbon binder phase was calculated using Eqs. 2.13, 2.14, respectively. If the calculated cell current was less than a specified tolerance of $1e-3$ then the solution was considered to be converged, otherwise a new value of potential was guessed at cathode current collector. Once the solution was converged, all field variables were updated and the simulation was run for the next time step. This was repeated until the cell voltage was greater than cut off voltage of 3 volts.

The pore network model was validated against experimental data for 1CAL and 3CAL electrodes reported by Usseglio-Viretta et al.²⁹ The cell potential vs capacity curves for 0.2, 0.5 and 1C rates of pore network model were compared with experimentally generated curves and found to agree well as shown in Fig. 4. Results of the pore network model at a discharge rate of 3C are also plotted for both calendared electrodes, even though experimental data were not available for this condition. The comparison between two electrodes at different levels of mechanical compaction is shown in Fig. 4c. It can be seen that the 3CAL electrode capacity fades more slowly than the 1CAL electrode, especially at higher discharge rates. This could be due to smaller cathode thickness and pore sizes in electrolyte phase and is discussed in detail below. It is also noteworthy that the 3CAL simulation matched the experiment data better than the 1CAL. One possible explanation for this is that thermal effects were not considered in the present model, and the better contact between the particles in the 3CAL case may have led to a more isothermal cell.

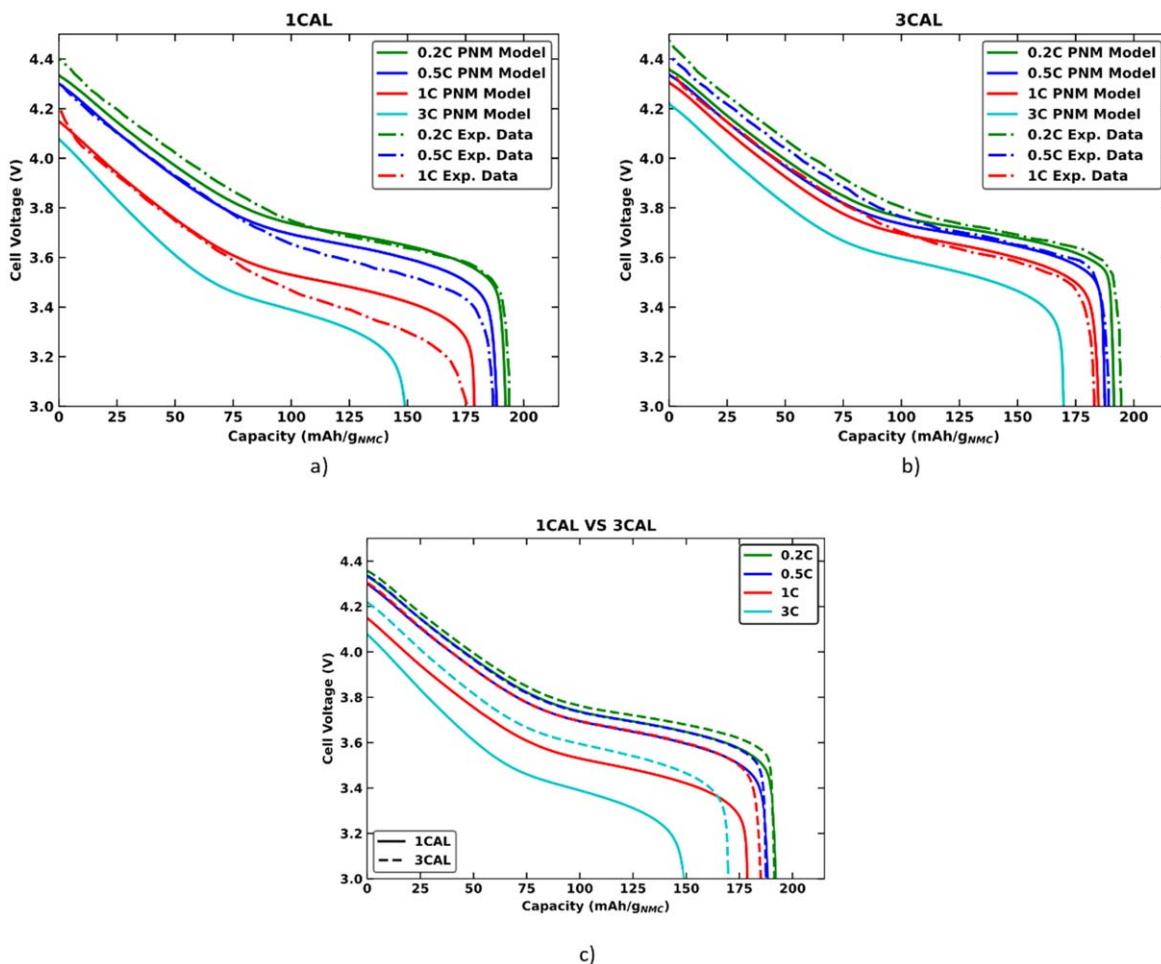


Figure 4. (a), (b) PNM and experimental voltage vs capacity curve for 1CAL and 3CAL respectively, (c) Comparison of voltage vs capacity curves at different rates for 1CAL and 3CAL.

Results and Discussion

Structural analysis of cathodes.—Figures 1e, 1m shows pore size distribution as a function of distance from the separator of 1CAL and 3CAL cathodes, respectively. The size of pores is indicated by its marker size and colour. Due to the additional calendaring applied to 3CAL electrodes, the overall size of the pores is smaller than the 1CAL cathode. The smaller size can be attributed to the impaction of some solid particle into the void space, decreasing the average diameter of pores. The larger pores can store more Li^+ ions for reaction inside the electrolyte phase. They also offer less electrolyte concentration gradient than small pores. Larger pores nearer to the separator/cathode interface can increase the mass transport of Li^+ ion in the cathode region which could ultimately increase the discharge capacity as reported in other studies,^{26,41} and hence increased intercalation/deintercalation of lithium-ion from electrolyte to the active material phase. Figures 1f, 1n shows distributions of particle sizes in the active material phase of 1CAL and 3CAL cathodes, respectively. The sizes of the active material particles are similar for both electrodes with only some differences in the spatial distribution of bigger particles as a function of separator distance. This is expected, given that calendaring is not expected to crack particles unless a large force is applied. Figures 1g, 1h shows the sphericity of pores and particles in both electrodes. Sphericity is defined as the ratio of surface area of a sphere having the same volume as the object to the actual surface area of the object.⁴² It can be seen that the sphericity of moderate to large sized particles is scattered in a narrow range as compared to the pores which are defined by the walls of the particles, so are concave

and tend to be less spherical. This trend is the same for both cathodes and is aligned with the structure of NMC532 in which particles are generally spherical. Figures 1h, 1p shows size distribution histograms of all three phases in 1CAL and 3CAL cathodes. It can be seen that CBD grains are smaller than the average pore and particle size. Also, the size of pores in 3CAL is smaller than 1CAL, which is also observed in Figures 1e, 1m. Of course, it must be remembered that the CBD phase was added artificially to these images, so this analysis is only approximate.

Besides size and shape distribution of cathodes, additional structural information can be obtained from the pore network extraction that is relevant to the transport processes. Figure 5a shows coordination number of each pore with its neighbouring pores, active material particles and CBD regions in both cathode samples. Higher connectivity generally means more transport paths, and therefore shorter transport lengths. It is thus desirable to maximize connectivity of the pore phase to improve Li -ion diffusion and migration in the electrolyte. Also, higher interconnectivity of pores-to-CBD can assist lithium-ion transport in CBD nanopores to reach the surface of the active material particles. On the other hand, Fig. 5b shows coordination numbers of active material particles with neighbouring particles, pores and CBD regions. The overall interconnectivity of active material particles is approximately the same in all phases indicating active material particles could structurally support both transport and reaction during the intercalation/deintercalation process. Figure 5c shows the slice-by-slice volume fraction of different phases in the tomography image of the cathode. The volume fraction in each case shows some variation which is due to the discrete nature of particles and pores. Figures 5d, 5e shows

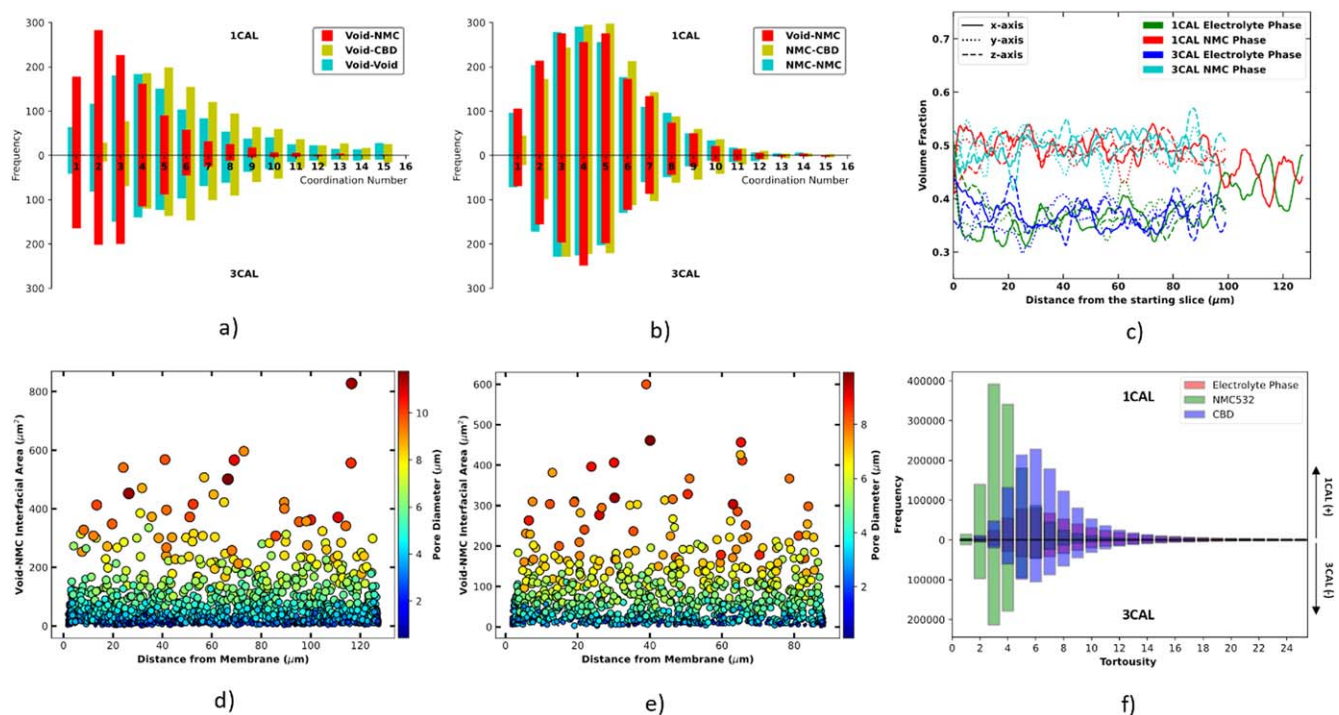


Figure 5. (a) Coordination number of pores with neighbouring pores, active material and CBD, (b) Coordination number of particles from neighbouring active material particles, pores and CBD, (c) Slice by slice variation of volume fraction of electrolyte and active material phase in different axis. The flow direction is along x -axis, (d), (e) shows interfacial area of active material accessible to each pore for 1CAL and 3CAL cathodes. This area is used in electrochemical reaction, (f) Local tortuosity of each pore, particle or CBD region, connected with every other possible pore, particle or CBD region respectively for both 1CAL and 3CAL cathode.

surface area of active material available to each pore in the electrolyte phase for 1CAL and 3CAL cathode samples respectively. The surface area of most of the particles is under $200 \mu\text{m}^2$ while a few pores with a relatively large size have more interfacial area available for reaction. Lastly, Figure 5f shows the network tortuosity, which was calculated by finding the shortest path between each node in a given phase to every other node in the same phase (found using the Dijkstra algorithm weighted by node-to-node distance), divided by the direct Euclidean distance between them. This network tortuosity was measured for all three phases and gives insights of interconnectivity of structure in each phase. Higher values mean the path between nodes is tortuous and it will offer high resistance to transport. This could lead to Li^+ exhaustion near the current collector specifically for thicker electrodes. It is therefore suggested that cathodes with higher thickness should have higher porosity to avoid resistance to diffusion of Li^+ to fully utilize the cathode material during the charging/discharging process. Ultimately, the properties mentioned in Fig. 5 give important information regarding microstructural heterogeneities inside porous cathode and they should be considered when optimizing electrode design during the manufacturing process or battery modelling.

Performance analysis of cathodes.—One of the benefits of pore-scale modeling is the ability to inspect the conditions in each pore and particle. Figure 6 shows discharge fields of both the 1CAL and 3CAL cathodes at rates of 1C and 3C at 75% state of lithiation (SoL), highlighting the spatial heterogeneities of the four computed transport variables in different phases of the cathode. To observe the effect of different variables like electrolyte and active material concentration and potential distribution during galvanostatic discharge, the diameter of pores and particles have been classified into three size zones or classes. In Figs. 1d, 1l the distribution of these zones in the electrolyte and active material can be seen for both electrodes. The diameter range of each zone is assigned a unique colour and will be used to study the effect of local heterogeneities on

the discharge behaviour. Figure 7 shows profiles of each variable for three different particle size ranges mentioned in Fig. 1d. The concentration decreases away from the separator as expected due to consumption of Li ions in the cathode. This behavior can of course be predicted by normal P2D models, but the distribution of concentrations at a given position can only be obtained with a pore-scale approach. For instance, the spatial distribution of concentration near the membrane is quite high ($100\text{--}200 \text{ mol/m}^3$), while the variation is much smaller near the current collector for all cases.

Figures 7e, 7k show electrolyte phase potential distribution of the 1CAL and 3CAL cathodes at 3C discharge rates. The potential drop in the electrolyte phase for the 3CAL electrode is much higher than the 1CAL sample, as expected due to higher current and concentration gradient during discharge at 3C rate. Also, as with the concentration profile, the spatial variation in potential distribution for the 3CAL electrode is low while on the other hand in the 1CAL electrode the potential has a broad distribution (approximately 200 mV at the current collector). This effect can be attributed to low concentration of Li^+ near current collector as seen in Fig. 6e. Due to fast electrochemical reaction, most of the Li^+ ions are consumed before reaching the current collector leading to deficiency of positive ions and hence increasing potential drop. Also, it can be seen that small pores respond to Li^+ deficiency faster than large pores because of inability to accumulate a significant number of Li^+ compared to larger pores. Although, deficiency of Li^+ due to concentration gradient can be studied using continuum models, the pore network approach enables identification of pores which specific pores will undergo Li^+ exhaustion earlier. This could help tailoring optimized structures for cathodes to more evenly distribute Li^+ throughout the porous structure for fast charging/discharging applications, for instance.

Figure 7 (right column) show the state of lithiation (SoL) in NMC active material. For the 1CAL electrode at 1C discharge (Fig. 7c) the SoL varies widely at all distances from the separator but the average value is relatively flat. It can also be observed that SoL is

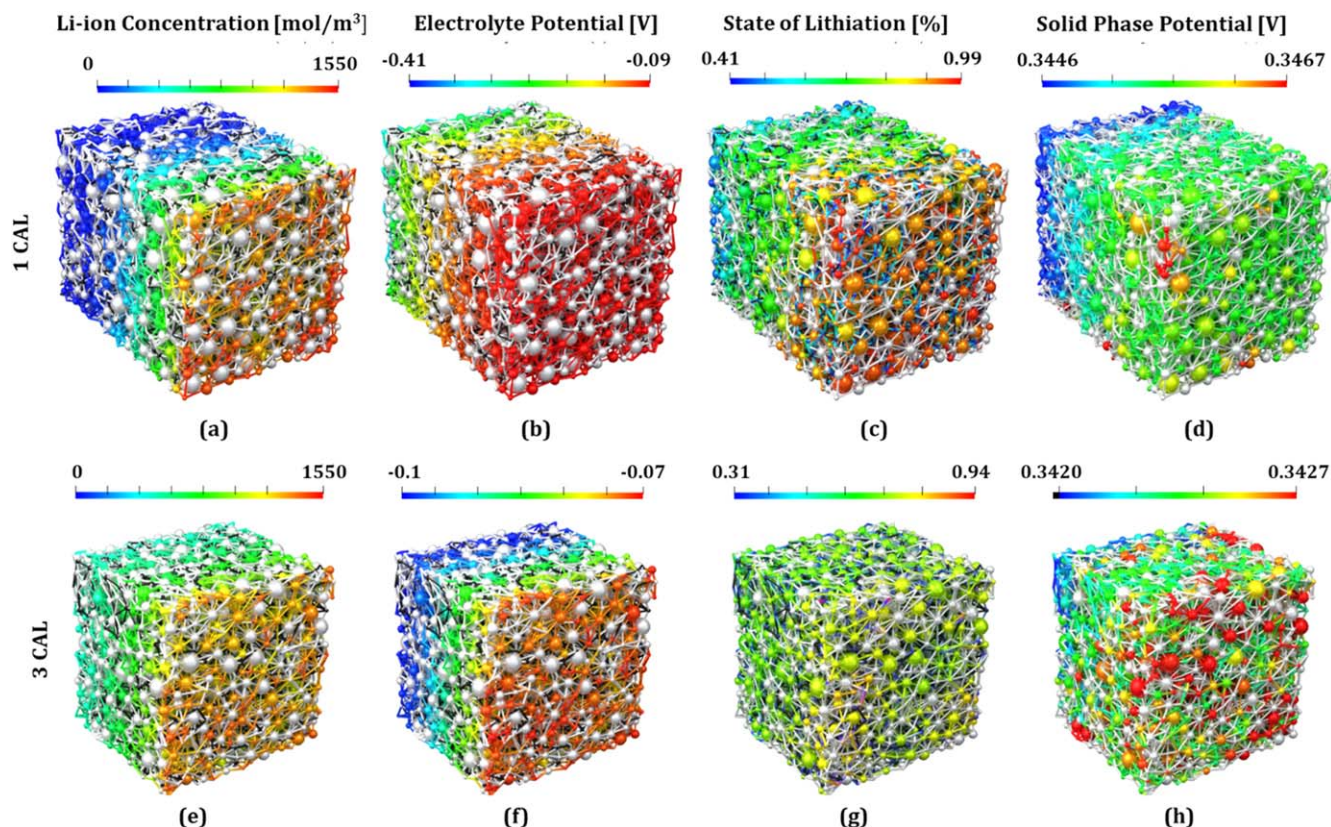


Figure 6. Spatial distributions of various solved quantities for C-rate of 3 in 1 CAL (top) and 3 CAL (bottom) cathodes. (a), (e) Li^+ concentration profile in electrolyte phase, (b), (f) Potential distribution in electrolyte phase, (c), (g) State of lithiation (SoL) spatial distribution in active material phase, (d), (h) Local variation of potential drop in NMC+CBD phase.

higher in smaller NMC particles compared to large particles as expected. At higher discharge rates the 1CAL electrode (Fig. 7f) still shows a wide variation of SoL, but there is also an overall gradient throughout the length of the cathode. The lithiation is almost completed near separator/cathode interface while it gradually decreases away from the separator. The low value of SoL can be attributed to reduced availability of Li^+ ions near the current collector in the electrolyte phase due to the long transport lengths through the domain. Also, smaller NMC particle show higher SoL indicating the importance of small size active material for fast charging of LIB's because of availability of more interfacial area as compared to bigger size particles to carry out intercalation reaction. In the 3CAL electrode (Figs. 7i, 7l) the SoL between particles varies much less, and there is a less pronounced drop across the domain. This is at least partly due to the shorter domain in the 3CAL electrode. The solid phase potentials are not shown since they vary by less than a few millivolts in all cases.

The spatial distribution of current density for 1CAL and 3CAL cathodes as a function of the distance from the membrane at 3C-rate is analyzed in Fig. 8, in terms of deviation from initial conditions. No experimental data is available for this case but given the good agreement of the model with the lower C-rates, it seemed justified to extend the analysis to higher current cases where transport effects are more pronounced. It was observed that for both cathodes the current density distribution in active material particle is more scattered at high SoC as compared to low SoC. Overall, the SoL in the 1CAL electrode showed more spatial distribution than in the 3CAL cathode. This could be due to higher thickness of 1CAL cathode adding more tortuous path for Li^+ and hence increasing resistance to diffusion. The particle-wise current density distribution also reveals an important insight about the possibility of electrode degradation. As seen in Fig. 8 the current density in some particles

are higher than their initial current density, especially near the cathode current collector. This behaviour could lead to rapid lithiation and subsequent crack of particles. This situation could be avoided by controlling the size of pores and particles especially near the cathode current collector side. In summary, the developed pore network modelling approach proved to be adept at predicting the performance of the electrode material. This is particularly impressive given that the only inputs into the model are a suitable image. It is foreseeable that next-generation materials and structures could be analyzed *in-silico* with this PNM approach to help screen and optimize proposed designs. Moreover, the pore-scale information provided by the PNM model can be used to predict degradation due to overly rapid lithiation of particles.

Computational performance.—Pore network modelling has the key advantage of performing simulations at vastly lower computational cost than the normal DNS approach. A steady-state performance comparison between the two approaches can be seen in our previous work.²² It should be mentioned that the network extraction step can be time consuming, but even very large images can be processed in less than 1 h.¹⁵ For the simulation of galvanostatic discharge at the different C-rates mentioned above two nested iterations loops were performed for each time step. The inner iteration to solve the Nernst Planck equation in electrolyte phase took about 1.25 seconds for 1C rate of 1CAL cathode consisting of 4637 nodes and 24799 bonds. While the outer iteration to converge the applied current density with calculated current density took 11.62 seconds for 1 time step. The tolerance values for inner and outer iteration were $1\text{e-}8$ and $1\text{e-}3$ respectively. For all simulations, a fixed time step of 2 seconds was used. Implementation of an adaptive time step scheme would offer additional performance improvements, but this is left for future work. The computation

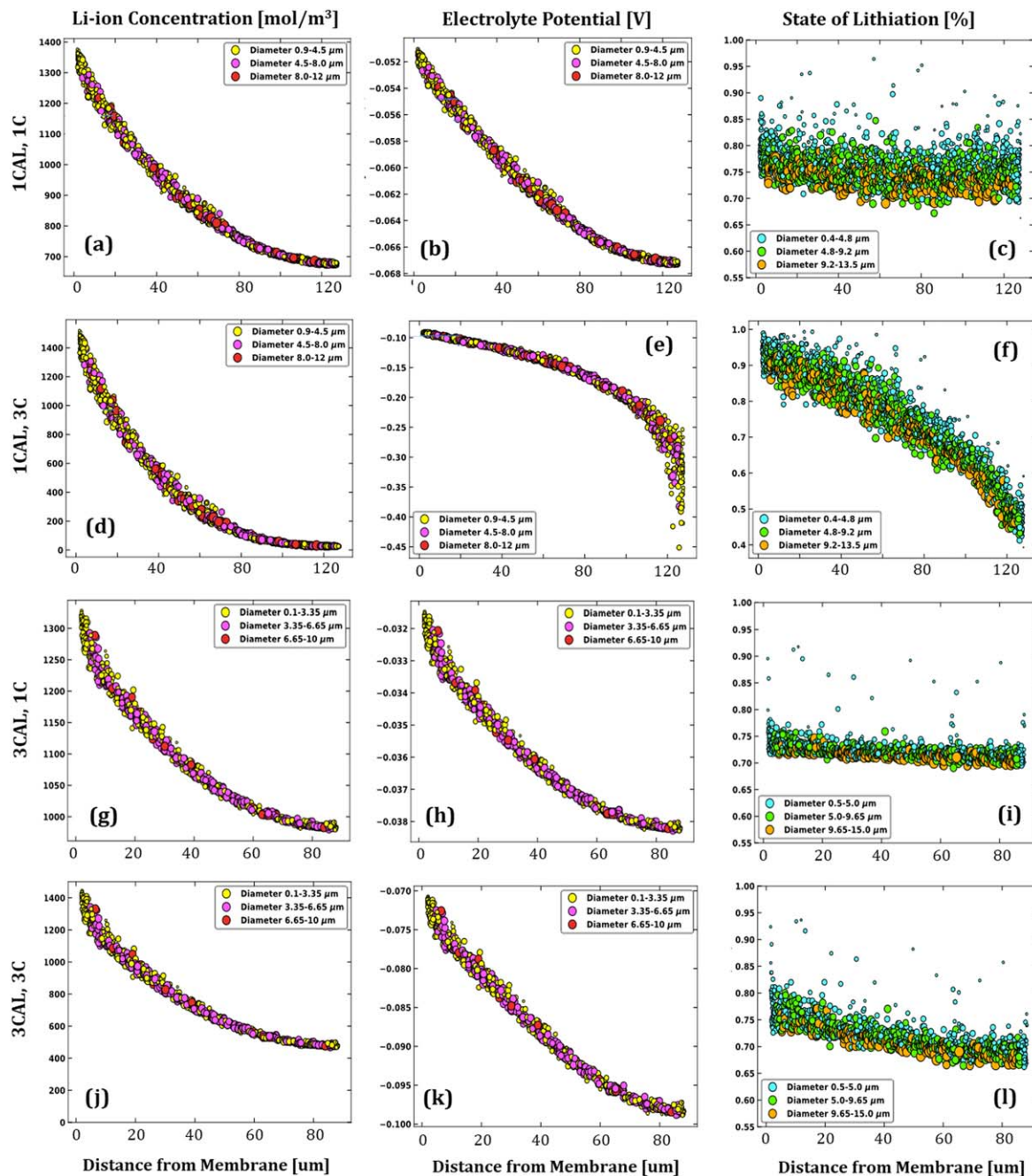


Figure 7. Profiles of solved quantities on a per-pore basis for 1CAL and 3CAL electrodes at discharge rates of 1C and 3C. Left column: Li^+ concentration profile as a function of separator distance, Center column: Spatial distribution of potential drop in electrolyte phase, Right column: State of lithiation (SoL). The markers are categorized in each figure according to the size class they belong to as indicated by the legend.

was carried using a Lenovo ThinkStation P720 workstation having an 8-core Intel(R) Xeon(R) Silver 4110 CPU @ 2.10GHz and 256 GB of RAM.

The overall computational performance time for the PNM was between 8 h for the 0.2 C case and 5 h for the 3 C. The 1 CAL electrode cases took slightly longer due to the higher number of nodes. The overall time increases as the C-rate decreases because more time steps were required. The 1 C case required about 6 h, meaning that the simulation was about 6x slower than real time. It is anticipated that PNMs could achieve real time simulations by employing adaptive time stepping or GPU-based solvers. The computational time reported for various DNS simulation is on the order of weeks; Hutzenlaub et al.⁴³ reported requiring 27 d to perform a simulation on the cluster using four nodes, where each

node contained two Intel Xeon E5-2680 CPUs and 32 GB RAM. Similarly, Rucci et al.⁴⁴ also reported simulation time up to 3 days depending on mesh size on a relatively small domain. Not only is this significantly more computing power than used in the present work, but the domain size used here is also 1.51 times bigger in terms of voxels. This computational performance gain of approximately 115X highlights the potential value of the pore network modelling approach to simulate complex multiphysics processes.

Conclusions

In this study, a pore network modelling framework was developed to analyze structure-performance relationship of two $\text{Li}(\text{Ni}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2})\text{O}_2$ cathodes of different thickness achieved by

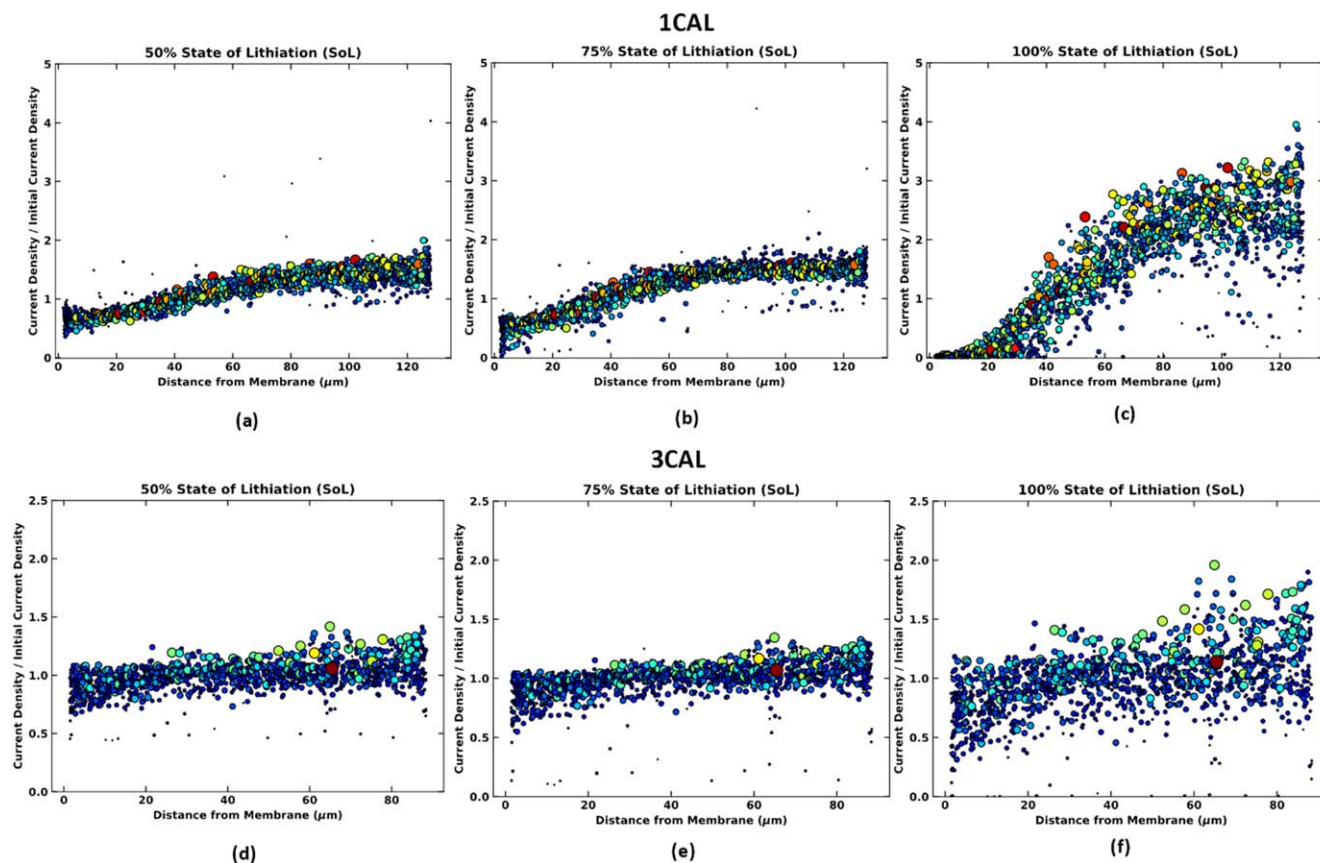


Figure 8. (a)–(c) Deviation of current density from initial current density at 50, 75 & 100% SoL respectively for 1CAL cathode, (d)–(f) Deviation of current density from initial current density at 50, 75 & 100% SoL respectively for 3CAL cathode. The color of circles indicates the size of active material particles with red and blue referring to big and small particles respectively.

differing calendaring pressures. A three-phase network was extracted from X-ray tomography images consisting of active material, carbon and binder domain and electrolyte phase along with interconnections of these phases with each other. The pore network modelling approach was used to obtain structural information about these two electrodes. It was confirmed that due to additional calendaring passes, the pore sizes in the 3CAL electrode were smaller than the 1CAL electrode. Sphericity data revealed that active material particles for both electrodes are similarly spherical, indicating that calendaring did not damage the particles, and that the particles were more spherical than electrolyte phase pores. Moreover, structural properties like coordination number, porosity, interfacial area and tortuosity of the extracted network revealed important information about local heterogeneities in porous electrodes.

The extracted networks were used to simulate galvanostatic discharge of lithium-ion battery with Li-foil anode to analyze the effect of local structural heterogeneities on four different transport variables in the cathode region. The pore network model was validated against experimental voltage-capacity curve data and found to agree well at a range of different C-rates. Ultimately, it was demonstrated that PNMs can efficiently predict the rate-dependent capacity of an electrode using only a 3-phase tomogram as input, which is a highly promising result. This suggests that *in-silico* structures that vary arbitrary properties of the electrode could be produced and screened for performance using PNMs, reduced the need to perform laborious and time-consuming cell fabrication and testing in the lab. Of course, this screening could also be performed using DNS, but the substantially improved computational performance of pore network modelling approach, at least $115\times$ faster compared to direct numerical simulation, is a key advantage. The developed pore numerical modelling framework provides a promising platform to optimize structural features of cathodes.

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